

JAMES PAOLO RILI

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SKILLS

Programming Languages	C, Rust, OCaml, MATLAB, Python, Java, Kotlin, TypeScript,
Libraries	NumPy, Matplotlib, Pandas, SciPy, scikit-learn, Keras
Frameworks	Compose, Express.js, React.js, Qt
Database	PostgreSQL, MongoDB, MySQL
Hardware	Electrical & RF Equipment, NI-VISA, Arduino, Raspberry Pi
Other Tools	Linux, Vim, AWS, Firebase, Git/GitHub, Atlassian, Microsoft Office

WORK EXPERIENCE

Software Developer Co-op, Reporting Services Caracal Technologies <i>Vancouver, BC</i>	May 2025 - Dec 2025
- Engineered new features for a web application processing high-volume transit data, delivering automated and accurate financial reporting for enterprise clients using Python via AWS Lambda for the backend and TypeScript with React.js on the frontend. - Refactored legacy modules and queries by applying functional and object-oriented principles, leading to increased system scalability and future maintainability. - Streamlined workflow in Bitbucket by writing Bash scripts and documenting standard procedures, reducing the risk of human error during deployment.	
RF Software Engineer Co-op NETGEAR, Inc. <i>Richmond, BC</i>	Sep 2022 - Apr 2023
- Developed and optimised Python-based software for WWAN validation of hotspot devices, reducing validation time by 50%. - Spearheaded the implementation of coding standards and led a comprehensive refactoring initiative, enhancing software scalability and maintainability. - Wrote documentation for users and developers of the software environment, reducing onboarding time from 6 to 3 days.	
Physics Teaching Assistant University of British Columbia <i>Vancouver, BC</i>	Jan - Apr 2024
- Conducted weekly workshops for 50 undergraduate students to reinforce understanding of complex problems. - Delivered concise lectures about waves and basic hydrostatics to ensure material comprehension. - Provided one-on-one assistance during office hours, assessing individual learning needs and reinforcing key concepts.	

EDUCATION

Computer Systems Technology Diploma	Sep 2024 - Present
British Columbia Institute of Technology	CGPA: 93%
- Functional Programming - System Design	- Databases - OOP - Operating Systems - Business Communications
Bachelor of Science in Physics	Sep 2019 - Apr 2024
University of British Columbia	
- Experimental Physics - Computational Physics	- Data Science - Software Construction - Statistical Mechanics - Writing for Research

PROJECTS

Iron-OBD2 BCIT + Iron IoT (https://github.com/ironiot/iron-obd2/)	Apr - May 2026
- Led a team of five in order to build a MVP of cloud-based OBD-II software, adding extended capabilities for vehicle diagnostics. - Communicated with client and teammates to ensure specifications are met and project on track to finish in under one month. - Wrote unit-testing with Python and automation code with Bash, accelerating deployment while maintaining bug-free software.	

- Built a custom database engine in C by implementing CRUD to handle student records via file I/O and memory management.
- Designed a command-line interface with the REPL pattern, leading to a modular codebase.
- Implemented error handling and input validation to ensure data integrity and system reliability.

EcoMeter | Enactus BCIT (<https://www.eco-meter.ca/>)

Sep 2024 - Apr 2025

- Assist the development team to maintain and optimise a student-run sustainable dining guide with Ruby on Rails.
- Integrate Mixpanel to provide user behaviour and engagement metrics for the club's data and marketing teams.
- Research and evaluate available technologies to identify opportunities for website performance and functionality improvements.

Machine Learning Interatomic Potentials | UBC (<https://arxiv.org/abs/2404.18393>)

Apr 2024

- Engineered a neural-network model with Keras API and object-oriented design to accurately predict crystalline atomic energies, providing a computationally efficient alternative to conventional, physics-based methods.
- Accelerated the search for the optimal model with feature engineering, cross-validation, and hyperparameter tuning.

Classification of Habitable Planets | UBC

Feb 2024

- Applied different machine learning classification algorithms to the NASA exoplanet dataset to determine most accurate model for determining habitable planets.
- Automated the model selection with Python through scikit-learn, along with NumPy and Pandas through Jupyter.

Characterisation of Semiconducting Materials | UBC (<https://arxiv.org/abs/2312.05744>)

Sep - Dec 2023

- Extracted material information of a semiconducting sheet with a specific four-point probe technique.
- Compiled a scientific paper on the results and its implications for future quantum computing hardware.
- Employed the use of NumPy and Pandas for data processing, analysis, and characterisation of semiconductor properties.

Numerical Simulations of Physical Phenomena | UBC

Jun 2023

- Wrote a program with MATLAB that simulates the physics of two colliding galaxies.
- Developed a physical model in Python that animates the behaviour of electron spin in a magnetic sheet.
- Designed and wrote simulations of 2-dimensional quantum particles in MATLAB to observe tunneling behaviour.

Circuit Prototyping Mini-Projects | Personal

Aug 2022

- Prototyped circuits with a breadboard, micro-controller, and sensing components to obtain data for physics experiments.
- Employed statistical techniques to reduce measurement uncertainty by 35%.
- Programmed the behaviour of components with the Arduino IDE and processed data with Python for fast data analysis.

Digit Recogniser | Personal

Jul 2022

- Built, trained, and tested a neural network that recognises handwritten digits in Python.
- Written with only the use of NumPy and linear algebra to understand fundamental concepts of neural networks.

Analysis of Johnson-Nyquist Noise in Electrical Components | UBC

Sep - Dec 2021

- Performed a series of experiments in an electrical laboratory investigating the effect of cosmic background noise on resistors.
- Analysed data using NumPy and Pandas, and configured calculations for theoretical models with SciPy.
- Composed a research paper detailing the process and presented the results to an audience of researchers and physicists.

ORGANISATIONS

UBC Physics Society

Sep 2019 - Apr 2024

General Member

Int'l Association of Physics Students

Sep 2022 - Apr 2024

General Member